

Slate v1.0 — AI-Native Pocket Workstation

Version 1.0 | Project: Slate Ecosystem — AI-Native Hardware Suite

Overview

Slate v1.0 is a 7.5-inch AI-native mobile workstation designed for full offline operation. It combines a high-density memory architecture, custom ARMv9 SoC, lateral camera system, and advanced passive cooling. Slate is built to run large language models (LLMs), Whisper, CLIP, and multimodal AI workloads without cloud dependency.

Component	Specification
Display	7.5" AMOLED, 3200×1800, 120Hz LTPO, 16:10
CPU	Custom ARMv9 12-core (4P + 4M + 4E)
GPU	8–10 TFLOPS FP16 RDNA3-class integrated GPU
NPU	100 TOPS transformer-optimized accelerator (INT4/INT8)
RAM	512GB LPDDR6X (stacked TSV interposer)
Storage	4TB LSSD (PCIe Gen5 x4, low-power SSD)
Cooling	Dual-zone vapor chamber, graphite spreaders, graphene frame conduction
Camera	48MP lateral optical system + under-display front camera
Battery	6,000mAh dual-cell, 100W USB-C PD, 50W wireless
Connectivity	Wi-Fi 7, Bluetooth 6, UWB 2.0, 5G optional
Chassis	Magnesium alloy + graphene-enhanced ceramic, IP68
Security	TPM 2.0, Secure Enclave, Airgap Mode, Encrypted RAM
OS	SlateOS — AI-native Linux-based system with local agent runtime
Weight	~250g Thickness: 8.8mm

Thermal & Structural Design

Slate employs a dual-zone vapor chamber with copper heat conduction and graphene spreaders. The lateral camera design redirects light through a prism into a horizontal lens array, allowing for a large 1/1.3-inch sensor without a rear bump. Phase-change material (PCM) beneath the NPU and RAM stabilizes temperatures under sustained AI workloads.

LLM & Multimodal Runtime

SlateOS integrates local runtime support for LLaMA 3 (13B–30B), Mixtral 8x7B, Phi-3, CLIP, and Whisper models. It supports quantization (INT4–BF16), context persistence, and vector memory for long-term reasoning. No cloud connection is required; all inference runs locally with encrypted model storage and session caching.

Security & Offline Operation

Slate prioritizes sovereignty and verifiable offline execution. It includes hardware TPM, RAM encryption, and a dedicated Secure Enclave for key management. Airgap Mode ensures total radio silence for critical environments.

Target Use Cases

• Offline LLM assistant and coding agent • Privacy-first creative workstation • Portable AI research tool for remote environments • Developer platform for on-device model testing and multimodal inference